

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10

1. A particle is moving in Simple Harmonic Motion in a straight line. In one minute of its motion it completes 15 oscillations and travels 120 metres. What is the amplitude of the motion?

- (A) 2 metres
- (B) 4 metres
- (C) 8 metres
- (D) 16 metres

2. If $z = \frac{1+2i}{3+4i}$, what is the modulus of z ?

- (A) $\frac{\sqrt{13}}{\sqrt{7}}$
- (B) $\frac{\sqrt{13}}{5}$
- (C) $\frac{\sqrt{5}}{5}$
- (D) $\frac{5}{\sqrt{5}}$

3. If $\frac{1}{x(x+1)} = \frac{a}{x} + \frac{b}{x+1}$, what are the values of a and b ?

(A) $a = -1, b = -1$

(B) $a = 1, b = -1$

(C) $a = -1, b = 1$

(D) $a = 1, b = 1$

4. Which of the following is a true statement about the lines l_1 and l_2 ?

$$l_1: \vec{r} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} \qquad l_2: y = -\frac{x}{2} + \frac{3}{2}$$

(A) l_1 and l_2 are parallel

(B) l_1 and l_2 are perpendicular

(C) l_1 and l_2 are neither parallel nor perpendicular

(D) l_1 and l_2 are the same line

5. Consider the following statement:

$\forall n \in \mathbb{N}, \text{ if } n^2 \text{ is odd, then } n \text{ is odd.}$

Which of the following is the negation of the statement?

(A) $\forall n \in \mathbb{N}, \text{ if } n^2 \text{ is odd, then } n \text{ is not odd}$

(B) $\exists n \in \mathbb{N}, \text{ if } n^2 \text{ is odd, then } n \text{ is not odd}$

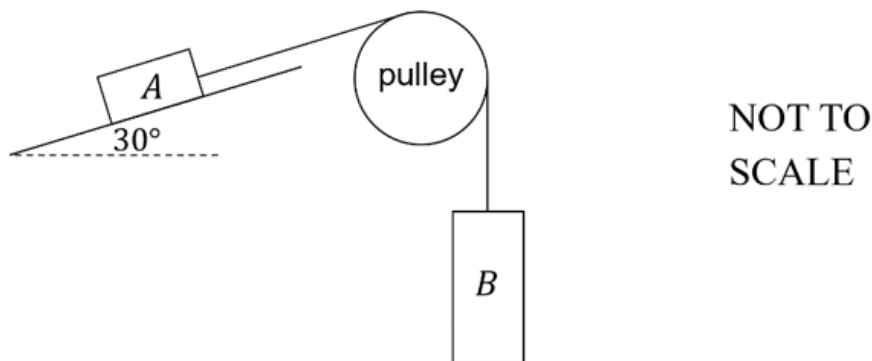
(C) $\forall n \in \mathbb{N}, \text{ if } n \text{ is not odd, then } n^2 \text{ is also not odd}$

(D) $\exists n \in \mathbb{N}, \text{ if } n \text{ is not odd, then } n^2 \text{ is also not odd}$

6. Consider the position vector of a particle $\vec{r} = -3\sin t \vec{i} + 3\cos t \vec{j} + t \vec{k}$.

Which of the following statements best describes the motion of the particle?

- (A) A helix about the z axis in an anticlockwise direction.
(B) A helix about the z axis in a clockwise direction.
(C) A helix about the x axis in an anticlockwise direction.
(D) A helix about the x axis in a clockwise direction.
7. Particles A and B , of masses M_A and M_B respectively, are connected by a light, inextensible string which passes over a frictionless pulley, as shown below:



Particle A lies on a frictionless surface inclined at 30° to the horizontal whilst particle B hangs vertically from the string.

The particles are initially at rest and, when they are released, neither particle moves.

Which expression shows the relationship between M_A and M_B ?

- (A) $M_A = M_B$
(B) $M_A = \sqrt{3}M_B$
(C) $M_A = 2M_B$
(D) $M_A = 2\sqrt{3}M_B$

8. Given the proposition:

$\forall n \in \mathbb{Z}$, if z^n is not purely imaginary, then z is not purely imaginary.

Given that each of the following statements is true, which statement disproves the proposition?

- (A) $(1 + i)^2 = 2i$
- (B) $(0 + i)^3 = -i$
- (C) $(1 + i)^3 = -2 + 2i$
- (D) $(0 + i)^4 = 1$

9. It is given that $f(a) = -2$, $f(b) = 8$, $g(a) = 1$ and $g(b) = 2$.

What is the value of $\int_a^b \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2} dx$?

- (A) 6
- (B) 2
- (C) -6
- (D) -18

10. Which of the following equations is equivalent to the intersection of

$|z| = |z - 2k - 2ki|$ and $|z - k - ki| < \sqrt{2}k$, where k is a positive constant?

- (A) $\arg(z - k) = \arg(z - ki)$
- (B) $\arg(z - k) = \arg(z - ki) + \pi$
- (C) $\arg(z - 2k) = \arg(z - 2ki)$
- (D) $\arg(z - 2k) = \arg(z - 2ki) + \pi$

End of Section I

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) (Start a new writing booklet.)

Marks

- a. On the Argand diagram provided in the Question 11 writing booklet sketch the region represented by the intersection of the inequalities: 3

$$1 \leq |z + 1| \leq 2 \quad \text{and} \quad \frac{\pi}{4} < \arg(z + 1) < \frac{2\pi}{3}$$

- b. Consider the vectors $\tilde{a} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$ and $\tilde{b} = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$

- i) Find $|\tilde{a}|$ 1

- ii) Find the angle between $\tilde{a}$ and $\tilde{b}$, in degrees correct to 1 decimal place. 2

- c. Find the equation of the vector which passes through the points (5, 4, 3) and (-2, 1, -1). 1

Question 11 continued on next page.

Question 11 continued.

d. Evaluate $\int_0^{\frac{\pi}{6}} \sin 3x \cos^4 3x \, dx$ 3

e. i) If $\frac{x^2+1}{x^3+2x^2+x} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$, find the values of A, B and C . 3

ii) Hence, find $\int \frac{x^2+1}{x^3+2x^2+x} \, dx$ 2

End of Question 11

Question 12 (15 marks) (Start a new writing booklet.)**Marks**

- a. $PQRS$ is a parallelogram with vertices $P(2, 4, 3)$, $Q(4, 3, 1)$, $R(3, 2, 5)$ and $S(x, y, z)$. 2

Find the coordinates of point S .

- b. A line l has vector equation $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix}$ 2

Show that the point $P(2, -5, 3)$ lies on line l .

- c. For $x \geq 0$, show that $\frac{x}{x^2+9} \leq \frac{1}{6}$ 2

- d. i) Given that $z = \cos\theta + i\sin\theta$, show that $z^n + z^{-n} = 2\cos n\theta$ 1

- ii) Hence, express $\cos^4\theta$ in terms of $\cos n\theta$. 3

- e. Using the substitution $u = \sin x + 1$, or otherwise, find 2

$$\int \frac{\cos x}{\sin^2 x + 2\sin x + 5} dx$$

Question 12 continued on next page.

Question 12 continued.

f. Consider the statement P in the set of integers:

If $5n^4 + 8n$ is a multiple of 16, then n is even.

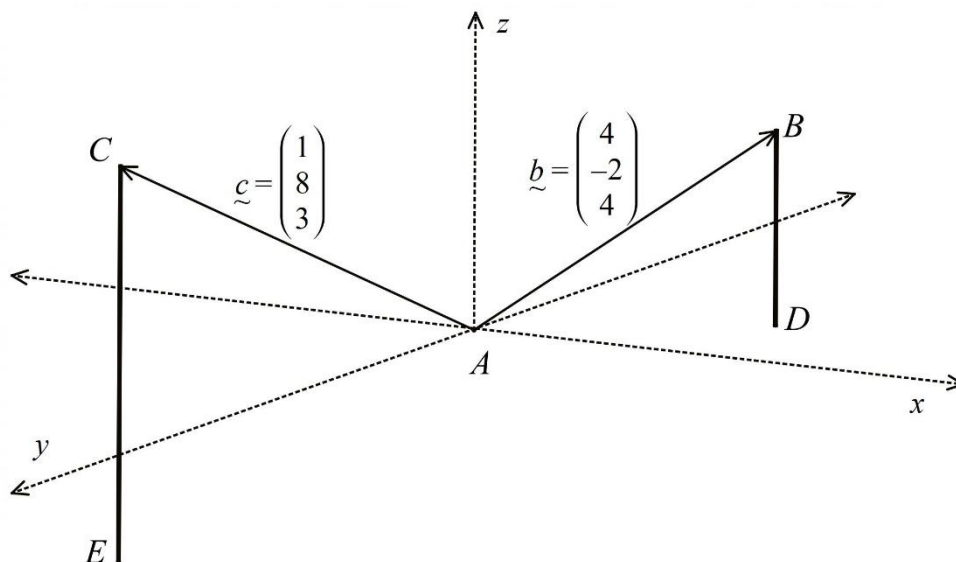
Prove that the converse of P is true.

3

End of Question 12

Question 13 (14 marks) (Start a new writing booklet.)**Marks**

- a. Two vertical posts CE and BD stand on a horizontal plane ADE . Using A as the origin, the vectors $\tilde{b}$ and $\tilde{c}$ represent the locations of the top of each post, B and C respectively. 2



Show that $\angle CAB$ is a right angle.

- b. Find $\int \frac{1}{\sqrt{6-5x-x^2}} dx$ 3

- c. Let l_1 be the straight line segment with equation 3

$$\tilde{r}_1 = \begin{pmatrix} 6 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ for } -3 \leq \lambda \leq 1.$$

The straight line segment l_2 is perpendicular to l_1 and the ends of l_2 are also its x and y intercepts.

Given that the endpoint of the position vector $\tilde{a} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ lies on both l_1 and l_2 , find the vector equation of l_2 .

Question 13 continued on next page.

Question 13 continued.

- d. Find the 4th roots of $1 + \sqrt{3}i$, leaving your answer in exponential form. 3

- e. Find the points of intersection of the line 3

$$\vec{r} = 3\vec{i} + 2\vec{j} - \vec{k} + \lambda(2\vec{i} - \vec{j} - 2\vec{k}) \text{ with the sphere}$$

$$(x - 3)^2 + (y - 2)^2 + (z + 1)^2 = 36$$

End of Question 13

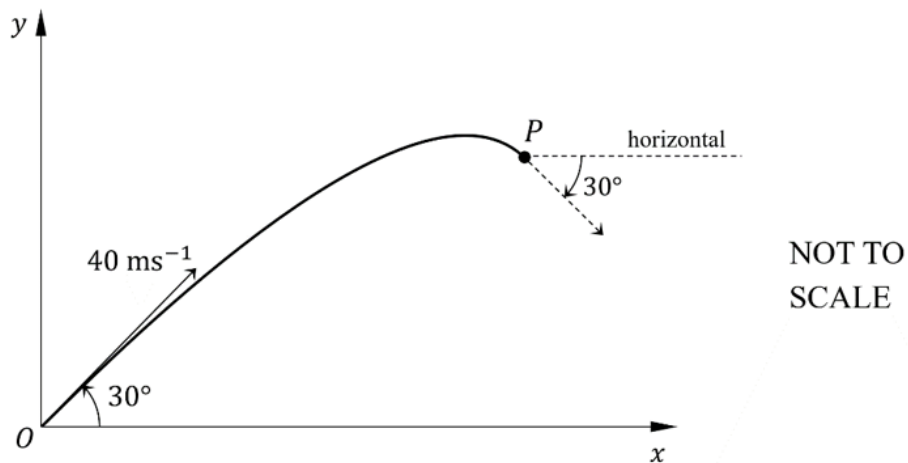
Question 14 (15 marks) (Start a new writing booklet.)**Marks**

- a. A particle of mass 1 kilogram is projected from the origin with speed 40 m/s at an angle of 30° to the horizontal plane as shown in the diagram for part ii) below.

The position vector of the particle at time t seconds after the particle was projected is given by:

$$\tilde{r}(t) = \begin{pmatrix} 5\sqrt{3}(1 - e^{-4t}) \\ \frac{45}{8}(1 - e^{-4t}) - \frac{5}{2}t \end{pmatrix} \quad (\text{Do NOT prove this})$$

- i) Find the velocity, $\tilde{v}(t)$, of the particle t seconds after it was projected. 1
- ii) At point P the particle is descending at 30° to the horizontal. 2



Show that the time taken for the particle to reach P is $\log_e \sqrt[4]{17}$ seconds.

Question 14 continued on next page.

Question 14 continued.

- b. The equations of two lines l_1 and l_2 are: 2

$$\underset{\sim}{r}_1 = \begin{pmatrix} -1 \\ -5 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2^k \\ 6 - 2^k \\ 7 - 2^k \end{pmatrix} \quad \text{and} \quad \underset{\sim}{r}_2 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 8 + 2^k \\ 6 \\ 5 + 2^k \end{pmatrix}$$

where λ is a real number and k is a constant.

For what value of k are the two lines parallel?

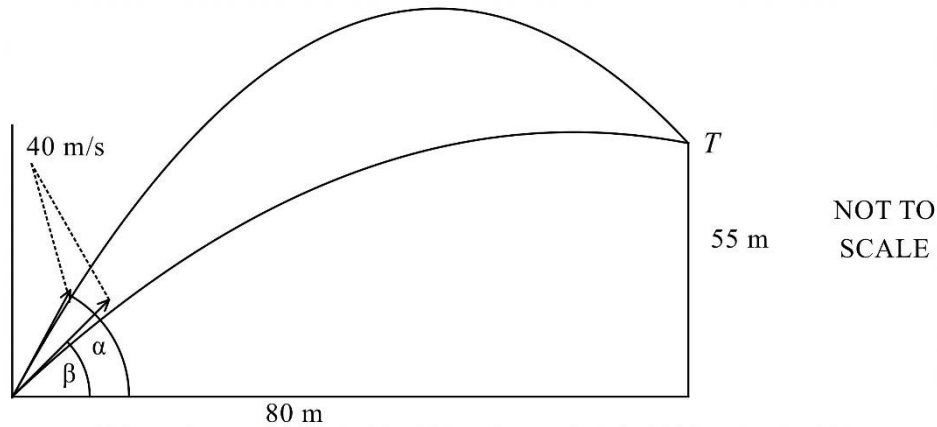
- c. The polynomial $P(z) = z^4 - 8z^3 + pz^2 + qz - 80$ has root $3 + i$, where p and q are real numbers.

- i) Find all of the roots of $P(z)$. 3
- ii) Write $P(z)$ as a product of real linear and quadratic factors. 1

Question 14 continued on next page.

Question 14 continued.

- d. A missile is fired into the air with velocity of 40 m/s at an angle of α above the horizontal. Shortly afterward a second missile is fired from the same point with the same velocity but at an angle of β above the horizontal, as shown below.



The missiles both simultaneously hit their target, T , which is 80m horizontally and 55 m vertically from the launch point.

The equations of motion for the first particle are given by:

$$x = 40t\cos\alpha \quad \text{and} \quad y = 40t\sin\alpha - \frac{gt^2}{2}$$

- i) Show that the equations of the paths of the missiles are given by:

$$y = x\tan\alpha - \frac{gx^2}{3200}\sec^2\alpha \quad \text{and} \quad 2$$

$$y = x\tan\beta - \frac{gx^2}{3200}\sec^2\beta \quad \text{respectively.}$$

- ii) Use $g = 10 \text{ m/s}^2$ to show that $\tan\alpha = \frac{5}{2}$ and $\tan\beta = \frac{3}{2}$ 2

- iii) Find the time, in exact form, which elapses between the firing of the two missiles. 2

End of Question 14

Question 15 (15 marks) (Start a new writing booklet.)**Marks**

- a. Consider the integrals: 2

$$I_n = \int_0^1 x^2(1 - x^2)^n dx \quad \text{and} \quad J_n = \int_0^1 (1 - x^2)^n dx$$

Show that $I_n = \frac{1}{2(n+1)} J_{n+1}$

- b. A body of mass one kilogram is projected vertically upwards from the ground at a speed of 40 m/s. The body is under the effects of both gravity and an air resistance which, at any time, has a magnitude of $\frac{v^2}{20}$, where v is the magnitude of the body's velocity at that time. Use $g = 10 \text{ m/s}^2$.

- i) Show that, while the body is travelling upwards, its equation of motion is: 1

$$\ddot{x} = -\left(10 + \frac{v^2}{20}\right)$$

- ii) Calculate the greatest height, in exact form, reached by the body. 2

- iii) Having reached its greatest height, the body falls back to the ground. The body remains under the effects of the same air resistance and gravity. 3

Show that the body is travelling at only one third of its initial launch speed when it hits the ground.

Question 15 continued on next page.

Question 15 continued.

c. i) Show that $\frac{2k}{k+2} < \frac{2k+2}{k+3}$ for $k > 0$ 1

ii) Use mathematical induction to prove that: 3

$$\frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \cdots + \frac{n}{(n+2)!} < \frac{2n}{n+2} - \frac{1}{(n+2)!} \text{ for } n \geq 1$$

d. A particle of mass M kilograms is moving up a slope which is at an angle of 27° to the horizontal. 3

The particle has initial velocity of u m/s up the slope and comes to a stop after 6 seconds.

It is subject to acceleration due to gravity of 10 m/s^2 and a resistive force due to friction of $0.2vM$ Newtons, where v is the velocity of the particle in m/s.

Find the value of u , correct to 2 decimal places.

End of Question 15

Question 16 (16 marks) (Start a new writing booklet.)**Marks**

- a. Prove by contradiction that, for $a \geq 2$: 3

$$\sqrt{a} + \sqrt{a+2} > \sqrt{a+8}$$

- b. Find $\int \frac{(2\tan\theta+3)\sec^2\theta}{\sec^2\theta+\tan\theta} d\theta$ 3

- c. It is known that the arithmetic mean of a set of numbers is greater than the geometric mean of the numbers.

That is:
$$\frac{x_1+x_2+x_3+\dots+x_n}{n} \geq \sqrt[n]{x_1x_2x_3 \dots x_n}$$

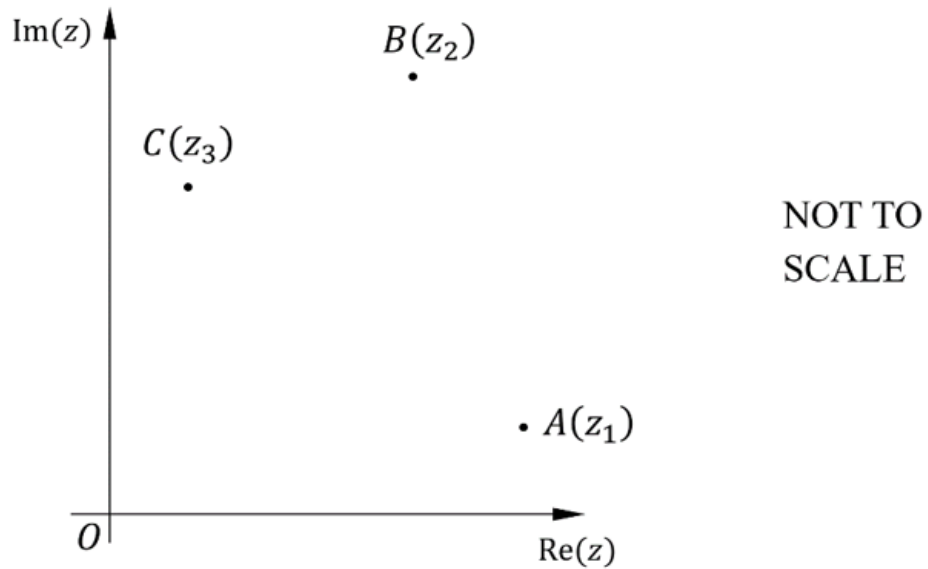
- i) Given that $a + b + c = k$ and $a, b, c > 0$, show that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{k}$ 3

- ii) Hence, given that $k = 1$, prove that $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) \geq 8$ 3

Question 16 continued on next page.

Question 16 continued.

- d. The complex numbers z_1, z_2 and z_3 are represented by the points A, B and C respectively. 4



The complex numbers satisfy $z_2 - z_1 = iz_1$ and $z_3 = e^{i\frac{\pi}{3}}z_1$

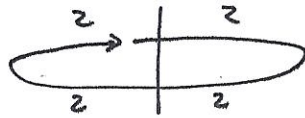
By considering $\angle BOC$ prove that $\cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{2\sqrt{2}}$

End of Question 16

End of Paper

2024 Ext 2 Trial Solns

1. $\frac{120}{15} = 8$

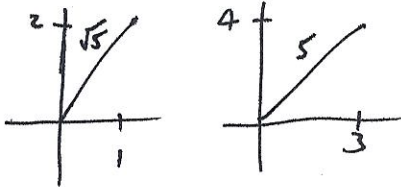


A

ACBBB

ACDAD

2.



$$|z| = \frac{\sqrt{5}}{5}$$

C

3.

$$\frac{1}{x(x+1)} = \frac{x+1}{(x)} \times \frac{-x}{(x+1)}$$

$$a = 1, b = -1$$

B

4.

$$m_1 = 2$$



$$m_2 = -\frac{1}{2}$$

B

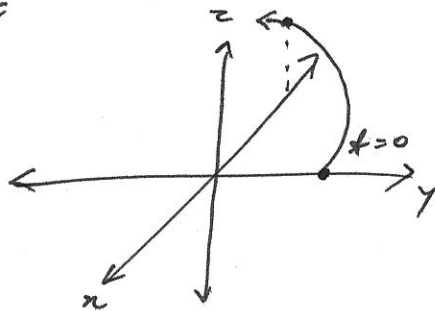
5.

$\forall n \in \mathbb{N}$, if P then Q
negates to

$\exists n \in \mathbb{N}$, if P then not Q

B

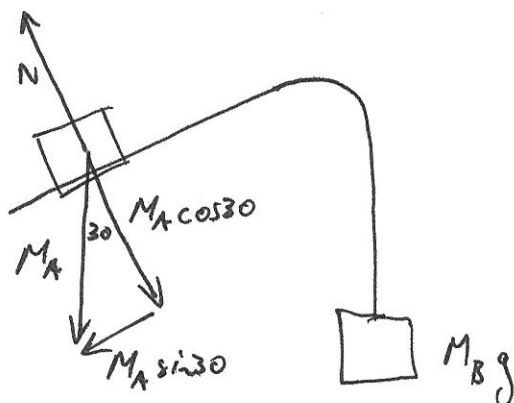
6. $\phi=0, \quad \underline{r} = 0\hat{i} + 3\hat{j} + 0\hat{k}$



$\phi = \frac{\pi}{2}, \quad \underline{r} = -3\hat{i} + 0\hat{j} + \frac{\pi}{2}\hat{k}$

A

7.



$$M_A \sin 30 = M_B$$

$$\frac{M_A}{2} = M_B$$

$$M_A = 2M_B \quad C$$

8. By obs $\Rightarrow D$ using contrapositive.

9.
$$\int_a^b \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} dx = \left[\frac{f(x)}{g(x)} \right]_a^b$$

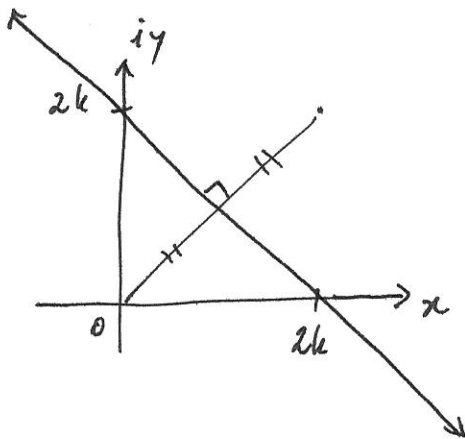
$$= \frac{f(b)}{g(b)} - \frac{f(a)}{g(a)}$$

$$= \frac{8}{2} - \frac{-2}{1}$$

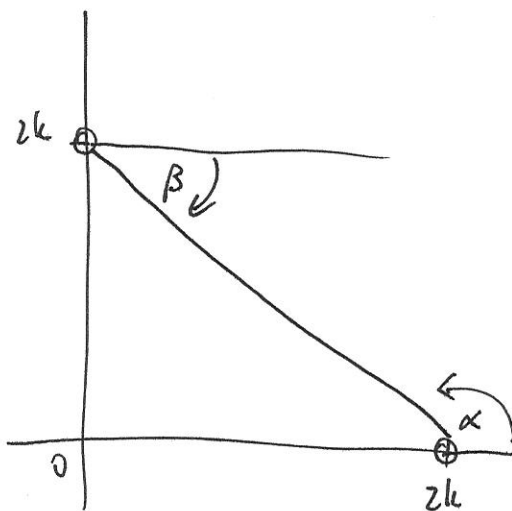
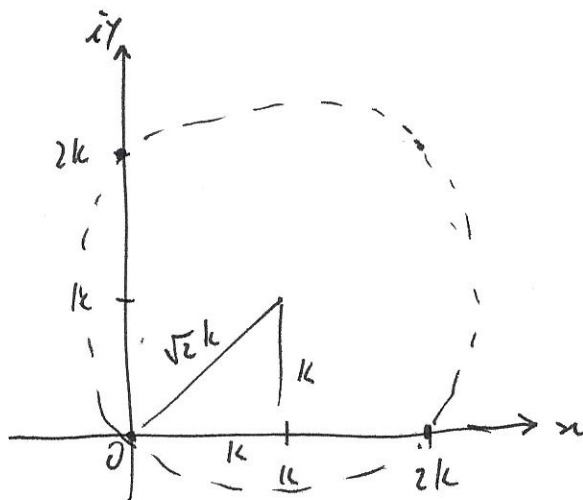
$$= 6$$

A

10. $|z| = |z - (2k + 2ki)|$



$$|z - (k + ki)| \leq \sqrt{2}k$$



$$\alpha - \beta = \pi$$

ie $\arg(z - 2k) - \arg(z - 2ki) = \pi$

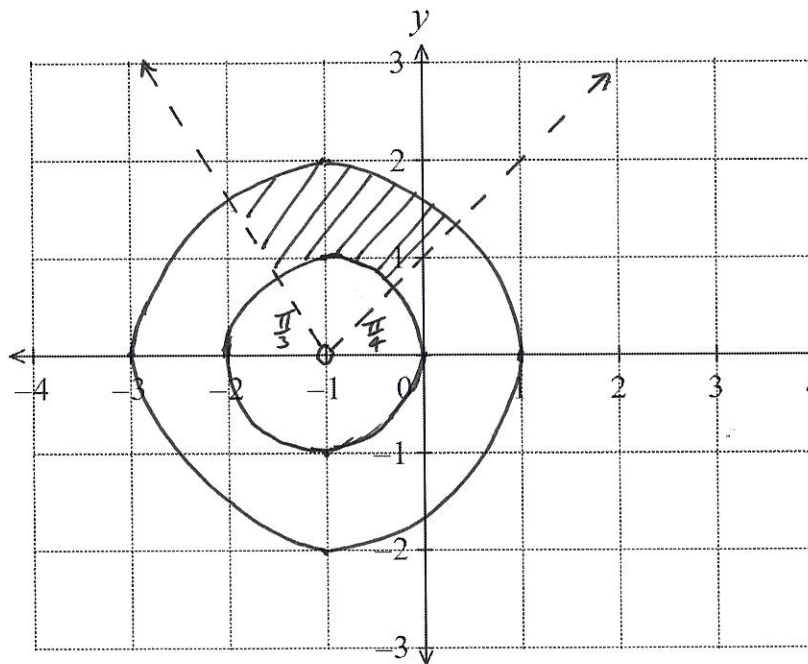
D

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Student Number

Start Here:

a. Use the diagram below for your answer to Question 11a.



① one correct

② minor error

③ correct.

$$b. i) \quad |a| = \sqrt{2^2 + 3^2 + 1^2} \\ = \sqrt{14}$$

①

$$ii) \quad |b| = \sqrt{(-1)^2 + 1^2 + 4^2} \\ = \sqrt{18} \\ = 3\sqrt{2}$$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

$$\therefore \cos \theta = \frac{2 \times -1 + 3 \times 1 + 1 \times 4}{\sqrt{14} \times 3\sqrt{2}}$$

① dot product.

$$= 0.3149703942$$

$$\theta = 71.64097347$$

$$= 71.6^\circ \text{ (1dp)}$$

①

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Student Number

c. $\underline{x} = \begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix} + 1 \begin{pmatrix} -7 \\ -3 \\ -4 \end{pmatrix}$ or other correct (1)

d. $\int_0^{\frac{\pi}{6}} \sin 3x (\cos 3x)^4 dx = -\frac{1}{3} \int_0^{\frac{\pi}{6}} -3 \sin 3x (\cos 3x)^4 dx$ (1)

$$= -\frac{1}{3} \left[\frac{(\cos 3x)^5}{5} \right]_0^{\frac{\pi}{6}} \quad (1)$$

$$= -\frac{1}{3} \left[\frac{\cos^5(\frac{\pi}{2})}{5} - \frac{\cos^5 0}{5} \right]$$

$$= -\frac{1}{3} \left(0 - \frac{1}{5} \right)$$

$$= \frac{1}{15} \quad (1)$$

e. 1) $\frac{x^2+1}{x^3+2x^2+x} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$

$$x^2+1 = A(x^2+2x+1) + Bx(x+1) + Cx \quad (1)$$

when $x=0$, $1 = A$ (1)

$x=-1$, $2 = -C$

$$C = -2$$

$x=1$, $2 = 4A + 2B + C$

$$2 = 4 + 2B - 2$$

$$2B = 0$$

$$B = 0$$

or other correct method.

ie. $A=1$, $B=0$, $C=-2$

(1) Additional writing space on the back page

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Student Number

$$\begin{aligned}
 11) \int \frac{x^2 + 1}{x^3 + 2x^2 + x} dx &= \int \frac{1}{x} - \frac{2}{(x+1)^2} dx \\
 &= \ln|x| - 2 \int (x+1)^{-2} dx \quad (1) \\
 &= \ln|x| - 2 \frac{(x+1)^{-1}}{-1} + C \\
 &= \ln|x| + \frac{2}{x+1} + C \quad (1)
 \end{aligned}$$

☐


Tick this box if you have continued this answer in another writing booklet

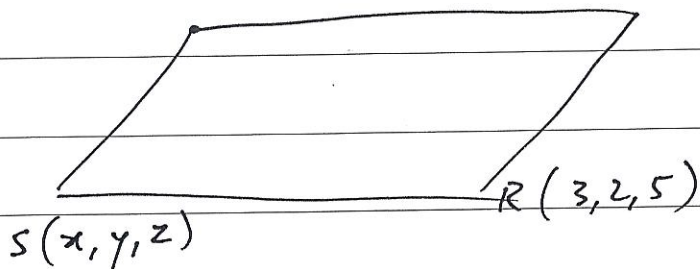
Start here for
Question Number:

12

$P(2, 4, 3)$

$Q(4, 3, 1)$

a.



① some sensible method.

$$\begin{aligned} x &= 3 - 2 \\ &= 1 \end{aligned}$$

$$\begin{aligned} y &= 2 + 1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} z &= 5 + 2 \\ &= 7 \end{aligned}$$

$\therefore S$ is $(1, 3, 7)$

①

b.

$$3 + 2\lambda = 2$$

$$2\lambda = -1$$

$$\lambda = -\frac{1}{2}$$

① find λ

Confirm that $\lambda = -\frac{1}{2}$ works for y, z coords

$$\begin{aligned} -2 + \frac{-1}{2} \times 6 &= -2 - 3 \\ &= -5 \end{aligned}$$

$\therefore$ correct for y coord

① confirmation
or, find λ
for all 3.

$$\begin{aligned} 5 - \frac{1}{2} \times 4 &= 5 - 2 \\ &= 3 \end{aligned}$$

$\therefore$ correct for z coord

$\therefore (2, -5, 3)$ lies on line l .

c. Consider $\frac{x}{x^2+9} - \frac{1}{6} = \frac{6x - (x^2+9)}{6(x^2+9)} \quad (1)$

$$= \frac{6x - x^2 - 9}{6(x^2+9)}$$

$$= \frac{-(x^2 - 6x + 9)}{6(x^2+9)}$$

$$= \frac{-(x-3)^2}{6(x^2+9)}$$

$$\leq 0 \quad \text{since } (x-3)^2 \geq 0 \quad (1)$$

and $6(x^2+9) > 0$

ie $\frac{x}{x^2+9} - \frac{1}{6} \leq 0$

$$\therefore \frac{x}{x^2+9} \leq \frac{1}{6}$$

OR

$$(x-3)^2 \geq 0 \quad (1)$$

$$x^2 - 6x + 9 \geq 0$$

$$x^2 + 9 \geq 6x$$

$$\frac{1}{6} \geq \frac{x}{x^2+9} \quad \text{since } x^2+9 > 0 \quad (1)$$

d. 1) $z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \quad (1)$

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

$$= 2 \cos n\theta$$

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11) Consider $\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$ (1)

$$\text{i.e. } (2\cos\theta)^4 = \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6 \quad (1)$$

$$16 \cos^4 \theta = 2 \cos 4\theta + 4 \times 2 \cos 2\theta + 6$$

$$\cos^4 \theta = \frac{\cos 4\theta}{8} + \frac{\cos 2\theta}{2} + \frac{3}{8} \quad (1)$$

$$= \frac{\cos 40^\circ + 4 \cos 20^\circ + 3}{8}$$

$$e. \int \frac{\cos x}{\sin^2 x + 2\sin x + 5} dx = \int \frac{\cos x}{(\sin x + 1)^2 + 4} dx$$

Let $v = \sin x + 1$

$$\frac{du}{dx} = \cos x$$

$$dv = \cos x \, dx$$

$$\therefore I = \int \frac{1}{v^2 + 4} dv \quad (1)$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{v}{2} \right) + C$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{\sin x + 1}{2} \right) + c \quad (1)$$



← Tick this box if you have continued this answer in another writing booklet.



Start here for

Question Number:

12

f. If P then Q Converse: If Q then P

(1)

ie. if n is even then $5n^4 + 8n$ is a mult. of 16let $n = 2P$ since n is even, where $P \in \mathbb{J}$

$$\therefore 5n^4 + 8n = 5(2P)^4 + 8(2P) \quad (1)$$

$$= 5 \times 16P^4 + 16P$$

$$= 16(5P^4 + P)$$

$$= 16Q \quad \text{where } Q = 5P^4 + P \text{ and}$$

$$Q \in \mathbb{J} \text{ since } P \in \mathbb{J} \quad (1)$$

 $\therefore$ If n is even then $5n^4 + 8n$ is a multiple of 16

Start here for
Question Number:

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$$a. \quad \underline{b} \cdot \underline{c} = |\underline{b}| |\underline{c}| \cos \theta$$

$$\begin{aligned} \underline{b} \cdot \underline{c} &= 1 \times 4 + 8 \times -2 + 3 \times 4 \\ &= 4 - 16 + 12 \\ &= 0 \end{aligned} \quad (1)$$

$$\text{ie } |\underline{b}| |\underline{c}| \cos \theta = 0$$

$$\therefore \cos \theta = 0$$

$$\text{since } |\underline{b}|, |\underline{c}| \neq 0$$

$$\therefore \theta = \frac{\pi}{2}$$

(1)

ie $\angle CAB$ is a right angle

$$b. \quad \int \frac{1}{\sqrt{6-5x-x^2}} dx = \int \frac{1}{\sqrt{6-(x^2+5x+(\frac{5}{2})^2)+\frac{25}{4}}} dx \quad (1)$$

$$= \int \frac{1}{\sqrt{\frac{49}{4} - (x + \frac{5}{2})^2}} dx$$

$$= \int \frac{1}{\sqrt{(\frac{7}{2})^2 - (x + \frac{5}{2})^2}} dx \quad (1)$$

$$= \sin^{-1} \left(\frac{x + \frac{5}{2}}{\frac{7}{2}} \right) + c$$

$$= \sin^{-1} \left(\frac{2x+5}{7} \right) + c \quad (1)$$

c. since $l_1 \perp l_2$, direction vector of l_2 will be $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ ①

Also, l_2 must pass through $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$

$$\therefore l_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad ①$$

find endpoints:

$$\text{when } x=0, \quad 2+\mu=0$$

$$\mu = -2$$

$$y=0, \quad 2-2\mu=0$$

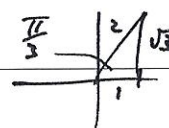
$$2\mu = 2$$

$$\mu = 1$$

$$\therefore l_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \text{for } -2 \leq \mu \leq 1 \quad ①$$

d. let $z^4 = 1 + \sqrt{3}i$

$$r^4 \text{cis } 4\theta = 2 \text{cis} \left(\frac{\pi}{3} \right) \quad ①$$



$$\therefore r^4 = 2 \quad \text{and} \quad 4\theta = \frac{\pi}{3} + 2k\pi, \quad k=0, \pm 1, \pm 2, \dots$$

$$r = \sqrt[4]{2}$$

$$= \frac{\pi + 6k\pi}{3}$$

$$\theta = \frac{\pi(1+6k)}{12} \quad ①$$

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when $k=0$, $z_1 = \sqrt[4]{2} e^{i\frac{\pi}{12}}$

spacing $\frac{2\pi}{4} = \frac{\pi}{2} = \frac{6\pi}{12}$

$$z_2 = \sqrt[4]{2} e^{i\frac{7\pi}{12}}$$

$$z_3 = \sqrt[4]{2} e^{-i\frac{11\pi}{12}}$$

$$z_4 = \sqrt[4]{2} e^{-i\frac{5\pi}{12}}$$

①

e. for $\underline{r}$, $x = 3 + 2\lambda$, $y = 2 - \lambda$, $z = -1 - 2\lambda$

$$\therefore (3 + 2\lambda - 3)^2 + (2 - \lambda - 2)^2 + (-1 - 2\lambda + 1)^2 = 36$$

①

$$4\lambda^2 + \lambda^2 + 4\lambda^2 = 36$$

$$9\lambda^2 = 36$$

$$\lambda = \pm 2$$

①

when $\lambda = 2$, $x = 7$, $y = 0$, $z = -5$
 $\lambda = -2$, $x = -1$, $y = 4$, $z = 3$

ie Pts of intersection are

$(7, 0, -5)$ and $(-1, 4, 3)$

①



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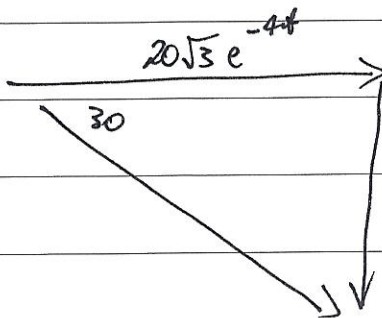


Start here for
Question Number:

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a. i)
$$\underline{r}(t) = \begin{pmatrix} 5\sqrt{3} - 5\sqrt{3}e^{-4t} \\ \frac{45}{8} - \frac{45}{8}e^{-4t} - \frac{5t}{2} \end{pmatrix}$$

$$\therefore \underline{v}(t) = \begin{pmatrix} 20\sqrt{3}e^{-4t} \\ \frac{45}{2}e^{-4t} - \frac{5}{2} \end{pmatrix} \quad (1)$$

ii) 

$$\therefore \tan 30 = \frac{\frac{5}{2} - \frac{45}{2}e^{-4t}}{20\sqrt{3}e^{-4t}} \quad (1)$$

$$\frac{20\sqrt{3}e^{-4t}}{\sqrt{3}} = \frac{5}{2} - \frac{45}{2}e^{-4t}$$

$$40e^{-4t} = 5 - 45e^{-4t}$$

$$85e^{-4t} = 5$$

$$e^{-4t} = \frac{1}{17}$$

$$-4t = \ln\left(\frac{1}{17}\right)$$

$$t = \frac{\ln\left(\frac{1}{17}\right)}{-4}$$

$$= -\frac{\ln(17)}{-4}$$

$$= \frac{1}{4} \ln(17)$$

$$= \ln(17)^{\frac{1}{4}}$$

$$= \ln(4\sqrt[4]{17})$$

(1)

b. l_1 and l_2 are parallel if direction vectors are in proportion.

$$\text{ie } \begin{pmatrix} 2^k \\ 6-2^k \\ 7-2^k \end{pmatrix} = c \begin{pmatrix} 8+2^k \\ 6 \\ 5+2^k \end{pmatrix} \quad \text{where } c \text{ is a constant} \quad (1)$$

$$\therefore 2^k = c(8+2^k) \quad \text{and} \quad 6-2^k = 6c$$

$$c = \frac{2^k}{8+2^k}$$

$$c = \frac{6-2^k}{6}$$

$$\therefore \frac{2^k}{8+2^k} = \frac{6-2^k}{6}$$

$$6 \times 2^k = 48 - 8 \times 2^k + 6 \times 2^k - (2^k)^2$$

$$(2^k)^2 + 8 \times 2^k - 48 = 0$$

$$(2^k - 4)(2^k + 12) = 0$$

$$\therefore 2^k = 4$$

or

$$2^k = -12$$

$$k = 2$$

which is not possible

$$\therefore k = 2 \text{ only}$$

(1)

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c. 1) Since coefficients are real, $3-i$ is also a root. (1)

Let the roots be $3+i$, $3-i$, α , β

$$\text{Now, } \alpha + \beta + 3+i + 3-i = 8$$

$$\alpha + \beta = 2$$

$$\text{and } \alpha\beta(3+i)(3-i) = -80$$

$$\alpha\beta(9+1) = -80$$

$$\alpha\beta = -8 \quad (1)$$

Form a quadratic in z with roots, α, β

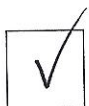
$$z^2 - 2z - 8 = 0$$

$$(z+2)(z-4) = 0$$

$$z = -2 \text{ or } z = 4$$

$\therefore$ The roots are $4, -2, 3+i, 3-i$ (1)

$$\begin{aligned} \text{ii) } \therefore P(z) &= (z+2)(z-4)(z-(3+i))(z-(3-i)) \\ &= (z+2)(z-4)(z^2-6z+10) \end{aligned} \quad (1)$$



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d. i) First missile:

$$t = \frac{x}{40 \cos \alpha}$$

$$\therefore y = -\frac{g}{2} \left(\frac{x^2}{1600 \cos^2 \alpha} \right) + 40 \sin \alpha \times \frac{x}{40 \cos \alpha} \quad (1)$$

$$= \frac{-gx^2}{3200 \cos^2 \alpha} + x \tan \alpha$$

$$= x \tan \alpha - \frac{gx^2}{3200} \sec^2 \alpha \quad (1)$$

Similarly, for 2nd missile

$$y = x \tan \beta - \frac{gx^2}{3200} \sec^2 \beta$$

ii) When missiles hit target, $x=80, y=55$

$$\therefore 55 = 80 \tan \alpha - \frac{10 \times 6400}{3200} (1 + \tan^2 \alpha) \quad (1)$$

$$55 = 80 \tan \alpha - 20 - 20 \tan^2 \alpha$$

$$20 \tan^2 \alpha - 80 \tan \alpha + 75 = 0$$

$$4 \tan^2 \alpha - 16 \tan \alpha + 15 = 0$$

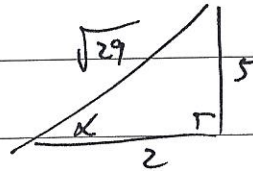
$$(2 \tan \alpha - 3)(2 \tan \alpha - 5) = 0$$

$$\tan \alpha = \frac{3}{2} \quad \text{or} \quad \tan \alpha = \frac{5}{2}$$

and, since $\alpha > \beta$

$$\tan \alpha = \frac{5}{2} \quad \text{and} \quad \tan \beta = \frac{3}{2} \quad (1)$$

III) $\tan \alpha = \frac{5}{2}$



$$\therefore \cos \alpha = \frac{2}{\sqrt{29}} \quad (1)$$

Similarly, $\cos \beta = \frac{2}{\sqrt{13}}$

Missile 1:

Time of flight

$$t = \frac{80}{40 \times \frac{2}{\sqrt{29}}}$$

$$= \sqrt{29}$$

Missile 2:

Time of flight

$$t = \frac{80}{40 \times \frac{2}{\sqrt{13}}}$$

$$= \sqrt{13}$$

$$\therefore \text{Time between firing is } \sqrt{29} - \sqrt{13} \text{ sec.} \quad (1)$$

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Start here for
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$$a. \quad I_n = \int_0^1 x^2 (1-x^2)^n dx$$

$$= -\frac{1}{2} \int_0^1 x \times -2x(1-x^2)^n dx$$

$$= -\frac{1}{2} \int_0^1 x \times \frac{d\left(\frac{(1-x^2)^{n+1}}{n+1}\right)}{dx} dx \quad (1)$$

$$= -\frac{1}{2} \left[\frac{x(1-x^2)^{n+1}}{n+1} \right]_0^1 + \frac{1}{2} \int_0^1 \frac{(1-x^2)^{n+1}}{n+1} dx$$

$$= 0 + \frac{1}{2(n+1)} \int_0^1 (1-x^2)^{n+1} dx \quad (1)$$

$$= \frac{1}{2(n+1)} J_{n+1}$$

OR

$$J_{n+1} = \int_0^1 (1-x^2)^{n+1} dx$$

$$= \int_0^1 \frac{d(x)}{dx} (1-x^2)^{n+1} dx \quad (1)$$

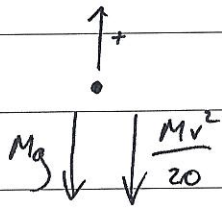
$$= \left[x(1-x^2)^{n+1} \right]_0^1 - \int_0^1 x(n+1)(1-x^2)^n \times -2x dx$$

$$= 0 + 2(n+1) \int_0^1 x^2 (1-x^2)^n dx$$

$$= 2(n+1) I_n \quad (1)$$

$$\therefore \frac{1}{2(n+1)} J_{n+1} = I_n$$

b. i)



$$\sum F = m\ddot{x} = -Mg - \frac{Mv^2}{20}$$

$$\therefore 1 \times \ddot{x} = -1 \times 10 - \frac{1v^2}{20}$$

$$\ddot{x} = -10 - \frac{v^2}{20} \quad (1)$$

$$= -\left(10 + \frac{v^2}{20}\right)$$

$$ii) \quad v \frac{dv}{dx} = -\left(\frac{200 + v^2}{20}\right)$$

$$\frac{dv}{dx} = \frac{-(200 + v^2)}{20v}$$

$$\frac{20v}{200 + v^2} dv = -dx$$

$$\int_{40}^0 \frac{2v}{200 + v^2} dv = \frac{-1}{10} \int_0^H dx \quad (1)$$

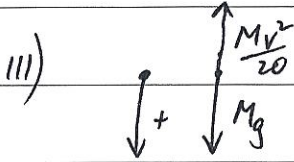
$$\left[\ln |200 + v^2| \right]_{40}^0 = \frac{-1}{10} \left[x \right]_0^H$$

$$\ln 200 - \ln 1800 = \frac{-1}{10} \times H$$

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$$\ln\left(\frac{1800}{200}\right) = \frac{H}{10}$$

$$H = 10 \ln 9 \quad (1)$$



$$\ddot{x} = 10 - \frac{v^2}{20}$$

$$v \frac{dv}{dx} = \frac{200 - v^2}{20} \quad (1)$$

$$\frac{20v}{200 - v^2} dv = dx$$

$$\int_0^v \frac{-2v}{200 - v^2} dv = -\frac{1}{10} \int_0^H dx$$

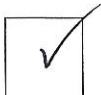
$$\left[\ln|200 - v^2| \right]_0^v = -\frac{1}{10} \left[x \right]_0^H \quad (1)$$

$$\ln|200 - v^2| - \ln 200 = -\frac{H}{10}$$

$$\ln|200 - v^2| = \ln 200 - \frac{10 \ln 9}{10}$$

$$= \ln\left(\frac{200}{9}\right)$$

$$\therefore 200 - v^2 = \frac{200}{9}$$



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Start here for

Question Number:

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$$v^2 = 200 - \frac{200}{9}$$

$$= \frac{1600}{9}$$

$$v = \frac{40}{3}$$

since $v > 0$ ①

ie hits ground at $\frac{1}{3}$ of launch speed.

c. 1) Consider $\frac{2k}{k+2} - \frac{2k+2}{k+3} = \frac{2k(k+3) - (2k+2)(k+2)}{(k+2)(k+3)}$

$$= \frac{2k^2 + 6k - (2k^2 + 6k + 4)}{(k+2)(k+3)}$$

$$= \frac{-4}{(k+2)(k+3)}$$

$$< 0 \quad \text{since } k > 0 \quad \text{①}$$

$$\text{ie } \frac{2k}{k+2} - \frac{2k+2}{k+3} < 0$$

$$\therefore \frac{2k}{k+2} < \frac{2k+2}{k+3}$$

ii) Prove true when $n=1$

$$\text{LHS} = \frac{1}{3!}$$

$$= \frac{1}{6}$$

$$\text{RHS} = \frac{2 \times 1}{1+2} - \frac{1}{(1+2)!}$$

$$= \frac{2}{3} - \frac{1}{6}$$

$$= \frac{1}{2}$$

$$\frac{1}{6} < \frac{1}{2}$$

①

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\text{ie. } \frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \dots + \frac{k}{(k+2)!} \leq \frac{2k}{k+2} - \frac{1}{(k+2)!}$$

Prove true for $n=k+1$

$$\text{ie. RTP: } \frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \dots + \frac{k}{(k+2)!} + \frac{(k+1)}{(k+3)!} < \frac{2k+2}{k+3} - \frac{1}{(k+3)!}$$

$$\text{LHS} = \frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \dots + \frac{k}{(k+2)!} + \frac{(k+1)}{(k+3)!}$$

$$< \frac{2k}{k+2} - \frac{1}{(k+2)!} + \frac{(k+1)}{(k+3)!} \quad \text{from assumption ①}$$

$$< \frac{2k}{k+2} + \frac{(k+1)}{(k+3)!} - \frac{1}{(k+2)!} \times \frac{k+3}{k+3}$$

$$< \frac{2k}{k+2} + \frac{k+1 - (k+3)}{(k+3)!}$$

$$< \frac{2k+2}{k+3} - \frac{2}{(k+3)!} \quad \text{from part (1)}$$

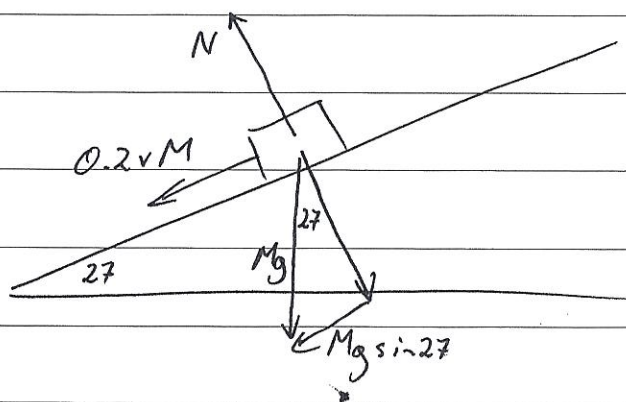
$$< \frac{2k+2}{k+3} - \frac{1}{(k+3)!} \quad \text{since } \frac{2}{(k+3)!} > \frac{1}{(k+3)!} \quad \text{①}$$

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$\therefore$ true for $n=k+1$

$\therefore$ by induction, the statement is true.

d.



In direction

of slope:

$$\sum F = M\ddot{x} = -0.2vM - Mg \sin 27$$

$$\ddot{x} = -0.2v - 10 \sin 27$$

$$\frac{dv}{dt} = \frac{-v}{5} - 10 \sin 27 \quad (1)$$

$$= - \frac{(v + 50 \sin 27)}{5}$$

$$\frac{dv}{v + 50 \sin 27} = \frac{-1}{5} dt$$

$$\int_0^0 \frac{1}{v + 50 \sin 27} dv = \frac{-1}{5} \int_0^6 dt$$

$$\left[\ln |v + 50 \sin 27| \right]_0^0 = \frac{-1}{5} \left[t \right]_0^6 \quad (1)$$



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Start here for

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$$\ln(50 \sin 27) - \ln(u + 50 \sin 27) = \frac{-6}{5}$$

$$\ln\left(\frac{u + 50 \sin 27}{50 \sin 27}\right) = \frac{6}{5}$$

$$\frac{u + 50 \sin 27}{50 \sin 27} = e^{\frac{6}{5}}$$

$$u = 50 \sin 27 \times e^{\frac{6}{5}} - 50 \sin 27$$

$$= 52.66555206$$

$$= 52.67 \text{ m/s (2dp)} \quad (1)$$

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a. Assume that

$$\sqrt{a} + \sqrt{a+2} \leq \sqrt{a+8}$$

$$(\sqrt{a} + \sqrt{a+2})^2 \leq a+8 \quad (1)$$

$$a + 2\sqrt{a^2+2a} + a+2 \leq a+8$$

$$2a + 2\sqrt{a^2+2a} + 2 \leq a+8$$

$$a + 2\sqrt{a^2+2a} \leq 6$$

Now, proof is for all $a \geq 2$ and smallest value of $a + 2\sqrt{a^2+2a}$ will occur when $a=2$ (1)

$$\text{When } a=2, \quad a + 2\sqrt{a^2+2a} = 2 + 4\sqrt{2}$$

$$\text{which leads to } 2 + 4\sqrt{2} \leq 6$$

$$4\sqrt{2} \leq 4$$

$$\sqrt{2} \leq 1 \quad (1)$$

which is a contradiction

$$\therefore \sqrt{a} + \sqrt{a+2} \not\leq \sqrt{a+8}$$

$$\therefore \sqrt{a} + \sqrt{a+2} > \sqrt{a+8}$$

$$b. \int \frac{(2\tan\theta + 3)\sec^2\theta}{\sec^2\theta + \tan\theta} d\theta$$

$$= \int \frac{2\tan\theta\sec^2\theta + 3\sec^2\theta}{\sec^2\theta + \tan\theta} d\theta$$

$$= \int \frac{2\tan\theta\sec^2\theta + \sec^2\theta}{\sec^2\theta + \tan\theta} d\theta + 2 \int \frac{\sec^2\theta}{\sec^2\theta + \tan\theta} d\theta \quad (1)$$

$$= \ln |\sec^2 \theta + \tan \theta| + 2 \int \frac{\sec^2 \theta}{\tan^2 \theta + \tan \theta + 1} d\theta$$

For $I = 2 \int \frac{\sec^2 \theta}{\tan^2 \theta + \tan \theta + 1} d\theta$ let $u = \tan \theta$
 $\frac{du}{d\theta} = \sec^2 \theta$

$$du = \sec^2 \theta d\theta$$

$$\therefore I = 2 \int \frac{1}{u^2 + u + 1} du \quad (1)$$

$$= 2 \int \frac{1}{(u^2 + u + \frac{1}{4}) + \frac{3}{4}} du$$

$$= 2 \int \frac{1}{(u + \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} du$$

$$= 2 \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{u + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C$$

$$= \frac{4\sqrt{3}}{3} \tan^{-1} \left(\frac{2u + 1}{\sqrt{3}} \right) + C$$

$$= \frac{4\sqrt{3}}{3} \tan^{-1} \left(\frac{2\tan \theta + 1}{\sqrt{3}} \right) + C$$

$$\therefore \int \frac{(2\tan \theta + 3)\sec^2 \theta}{\sec^2 \theta + \tan \theta} d\theta = \ln |\sec^2 \theta + \tan \theta| + \frac{4\sqrt{3}}{3} \tan^{-1} \left(\frac{2\tan \theta + 1}{\sqrt{3}} \right) + C \quad (1)$$

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c. i) Using AM-GM

$$\frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \sqrt[3]{\frac{1}{a} \times \frac{1}{b} \times \frac{1}{c}} \quad (1)$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{3}{\sqrt[3]{abc}}$$

Also, $\frac{k}{3} \geq \sqrt[3]{abc} \quad (1)$

$$\frac{3}{k} \leq \frac{1}{\sqrt[3]{abc}} \quad \text{since all terms are positive}$$

$$\frac{9}{k} \leq \frac{3}{\sqrt[3]{abc}} \quad (1)$$

$$\therefore \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{k}$$

ii) since $k=1$, $a+b+c=1$
and $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$

$$\begin{aligned} \text{Now, } \left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) &= \frac{(1-a)}{a} \times \frac{(1-b)}{b} \times \frac{(1-c)}{c} \quad (1) \\ &= \frac{(1-a)(1-b-c+bc)}{abc} \\ &= \frac{1-b-c+bc-a+ab+ca-abc}{abc} \end{aligned}$$



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$$= \frac{1 - (a+b+c) + (ab+bc+ca) - abc}{abc} \quad (1)$$

$$= \frac{1 - 1 + ab + bc + ca - abc}{abc}$$

$$= \frac{ab+bc+ca}{abc} - \frac{abc}{abc}$$

$$= \frac{1}{c} + \frac{1}{a} + \frac{1}{b} - 1$$

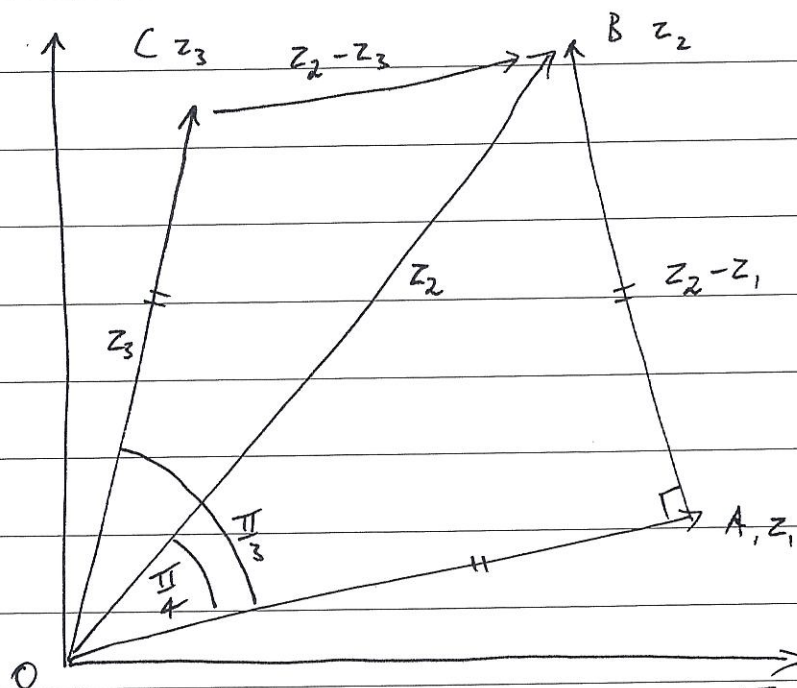
$$\geq 9 - 1$$

$$\text{since } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$$

(1)

$$\geq 8$$

d.



$$z_2 - z_1 = iz_1$$

$$z_2 = z_1 + iz_1$$

$$= (1+i)z_1$$

$$z_3 = e^{i\frac{\pi}{4}} z_1$$

$$= \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) z_1$$

$$= \left(\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) z_1$$

Since $z_2 - z_1 = iz_1$,

$$|z_2 - z_1| = |z_1|$$

and so $\triangle OAB$ is
right angled at A and
isosceles.

$$\therefore \angle AOB = \frac{\pi}{4}$$

Since $z_3 = e^{i\frac{\pi}{3}} z_1$,

$$\angle AOC = \frac{\pi}{3}$$

$$\text{and } |z_3| = |z_1|$$

①
either

$$\therefore \angle BOC = \frac{\pi}{3} - \frac{\pi}{4}$$

$$= \frac{\pi}{12}$$

①

Using the cosine rule in $\triangle OBC$

$$\cos \frac{\pi}{12} = \frac{|z_3|^2 + |z_2|^2 - |z_2 - z_3|^2}{2 \times |z_3| \times |z_2|}$$

$$= \frac{|z_1|^2 + (\sqrt{2}|z_1|)^2 - |z_2 - z_3|^2}{2 \times |z_1| \times \sqrt{2}|z_1|}$$

Now, $z_2 - z_3 = (1+i)z_1 - \left(\frac{1}{2} + \frac{\sqrt{3}i}{2}\right)z_1$

$$= \left(\frac{1}{2} + \left(1 - \frac{\sqrt{3}}{2}\right)i\right)z_1$$

①

$$|z_2 - z_3| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(1 - \frac{\sqrt{3}}{2}\right)^2} \times |z_1|$$

Additional writing space on back page.

$$= \sqrt{\frac{1}{4} + 1 - \sqrt{3} + \frac{3}{4}} \times |z_1|$$

$$= \sqrt{2 - \sqrt{3}} \times |z_1|$$

$$\therefore \cos \frac{\pi}{12} = \frac{|z_1|^2 [1 + 2 - (2 - \sqrt{3})]}{|z_1|^2 \times 2\sqrt{2}} \quad (1)$$

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$



⇐ Tick this box if you have continued this answer in another writing booklet.

